

Muhammad Rehan

Email | Website | GitHub | LinkedIn | Academia.edu

Education

- 2023– **Graduate Student**, Program in Indo-European Studies, UCLA
- 2023–2024 **M.A. in Indo-European Studies**, UCLA
- 2022–2023 **Post-Baccalaureate Certificate in Classics** (with Distinction), UCLA
- 2018–2022 **B.A. in Classics (Hons.) and German Studies (Hons.)**, Grinnell College

External Coursework and Certifications

- 2026 Advanced Bayesian Data Analysis, Ninth Summer School on Statistical Methods for Linguistics and Psychology, Potsdam
- 2025 Linguistic Typology in Evolutionary Perspective, Statistical Measures in Corpus Linguistics, Foundations of Predictive Modeling with R, Variationist Sociolinguistics, Reconstructing Syntax, Adaptive Speech Perception and Comprehension, Language Phylogenies, LSA Summer Institute, University of Oregon
- 2024 Exceptionality, Phonological Grammar and Opacity, Historical Linguistics, CreteLing 2024
- 2021 Old English, Leiden Summer School in Languages and Linguistics
- 2021 The Ancient Greek Workshop, UC Berkeley
- 2020 Deutsch Erleben Workshop, Humboldt Universität
- 2019 German Language Program, Universität Leipzig

Research Interests

Quantitative historical linguistics; probabilistic models of sound change; linguistic phylogenetics; Bayesian spatiotemporal modeling; Agent-based modeling; Indo-European philology and linguistics.

Publications

Peer-Reviewed Articles and Articles under Review

- Submitted* **Muhammad Rehan**, Valentina Lunardi, and David Goldstein, “**Supervised measurement of Christian Latin.**”
Digital Scholarship in the Humanities
- 2026 Grant S. McCall and **Muhammad Rehan**, “**E-lorgesailie: An agent-based simulation model of Acheulean lithic raw material procurement and handaxe deposition patterns.**”
Journal of Archaeological Method and Theory 33.49.
- 2021 Grant S. McCall, Russell Greaves, Brian Hitchcock, Brian Ostahowski, Sherman W. Horn III, and **Muhammad Rehan**, “**The estuarine ecological knowledge network: Future prospects.**”
Marine Technology and Science Journal 55 (3).122–123.

Conference Proceedings

- Forth.* **Muhammad Rehan**, “**Re-evaluating the word-medial operation of the Saussure-Hirt Effect in Greek.**”
Proceedings of the 4th International Conference on Ancient Greek Linguistics.
- 2026 **Muhammad Rehan**, “**Gk. μοιχός ‘adulterer’: Re-mixing a piss-poor etymology**”
Proceedings of the 35th Annual UCLA Indo-European Conference Vol. 35.147–163.

Manuscripts in Preparation

- In Prep.* **Muhammad Rehan** and David Goldstein, “**From points to polygons: Modeling spatial autocorrelation.**”
- In Prep.* **Muhammad Rehan** and Kie Zuraw, “**MUSE: An R package for ethnomusicological analysis.**”
- In Prep.* **Muhammad Rehan** and Megha Sundara, “**Young infants’ discrimination of phonetic contrasts predicts their typological frequency.**”

Other Publications

- 2023 **Muhammad Rehan**, “**Mind your language, Mister! A case study in Aristophanic humor.**” *Animus* Vol. 3.1 (Winter).36–51.
- 2023 **Muhammad Rehan**, “**Between need, want, material and innate beauty: Tracing shades of Erōs.**” *Corvus* Vol. 13 (Spring).27–35.
- 2023 **Muhammad Rehan**, “**Gendered language in Aristophanes’ *Birds*.**” *Persephone* Vol. 8.1 (Spring).12–29.

Presentations & Posters

Quantitative and Computational Historical Linguistics

- 2026 Tom Francis and **Muhammad Rehan**, “**Modeling alphabet diffusion and community interactions in ancient Italy with spatiotemporal point processes.**” *Connected Past* 2026, Sept. 22.
- 2026 **Muhammad Rehan** and David Goldstein, “**From points to polygons: Modeling spatial autocorrelation.**” *qc-leste: Quantitative Cross-Linguistic Explorations of Space, Time, and Evolution*, Jun. 10.
- 2026 Tom Francis and **Muhammad Rehan**, “**Point processes and spatiotemporal modeling of alphabet spread in Ancient Italy.**” *DataX, UCLA*, May 19.
- 2026 **Muhammad Rehan**, “**Why do languages they have the sounds they do: Insights from infant experiments and phylogenetic modeling.**” *DataX, UCLA*, April 20.
- 2026 **Muhammad Rehan**, “**Discerning segmental stability using phylogenetic models.**” *Southern California Annual Meeting on Phonology*, Apr. 11.
- 2026 **Muhammad Rehan** and David Goldstein, “**From points to polygons: Modeling spatial autocorrelation.**” *Special workshop on “The geography of linguistic evolution”, EVOLANG XVI*, Apr. 7.
- 2025 **Muhammad Rehan**, Sean Liu, and Marc Canby, “**Converging computational evidence against an East-Germanic clade.**” *WeCIEC 36: Annual UCLA Indo-European Conference*, Oct. 28.
- 2025 Tom Francis, Klaus Kim, Laura McLean, and **Muhammad Rehan**, “**Issues of lexical cognacy encoding in IE-CoR.**” *27th International Conference on Historical Linguistics*, Aug. 18.
- 2025 **Muhammad Rehan**, “**Population size has a weak effect on phoneme inventory size.**” Poster presented at the *Linguistic Society of America Summer Institute, University of Oregon*, Jul. 10.

- 2024 **Muhammad Rehan, “Towards an integrative approach to the investigation of linguistic phylogenies.”**
5th Munich-UCLA Historical Linguistics Colloquium, Jun. 28.
- 2024 **Muhammad Rehan, “FSAs, FSTs and sequence alignment in phylogenetic studies: The case of Tocharian B.”**
UCLA Program in Indo-European Studies Graduate Seminar, May 29.

Phonological Change and Indo-European Philology

- 2026 **Muhammad Rehan, “The phonetic value of Avestan <â>.”**
American Society for Premodern Asia Annual Meeting, Apr. 24.
- 2025 **Muhammad Rehan, “The development of word-final stops in Hittite.”**
AnatoLinks, University of Oxford, Dec. 4.
- 2025 **Muhammad Rehan, “Synchronic phonetic variation drives diachronic sound change: Further evidence from pre-nasal raising in Young Avestan.”**
7th Edinburgh Symposium on Historical Phonology, Dec. 2.
- 2025 **Muhammad Rehan, “Re-evaluating the word-medial operation of the Saussure-Hirt Effect in Greek.”**
11th International Colloquium on Ancient Greek Linguistics, June 26.
- 2025 **Muhammad Rehan, “Word-final voicing in Hittite?”**
Southern California Annual Meeting on Phonology 2025, Apr. 12.
- 2024 **Muhammad Rehan, “Re-mixing a piss-poor etymology: On Gk. μολχός and OE (ge)māh.”**
WeCIEC 35: Annual UCLA Indo-European Conference, Oct. 25.

Phonetics, Text-Setting, and Digital Philology

- 2026 Elisa Migliaretti, Spiridon-Iosif Capotos, **Muhammad Rehan**, and Zachary Fletcher, “MoPSoS: Introduction to the database of Ancient Greek morphology, prosody, syntax and scansion.”
“Epic through Numbers” panel, Celtic Conference in Classics, Jul. 15.
- 2026 **Muhammad Rehan, Jahnavi Narkar, and Dhanya Charan, “Lexical stress and syllable weight in Hindi-Urdu pop text-setting.”**
Linguistic Society of America Annual Meeting 2026, Jan. 9.
- 2025 **Muhammad Rehan, Jahnavi Narkar, and Dhanya Charan, “Investigating lexical stress in Hindi-Urdu pop music.”**
Sixth Joint Meeting of Acoustical Society of America and Acoustical Society of Japan, Dec. 5.

Guest Lectures

- 2026 **“How can experimental psycholinguistics and diachronic phonology help us learn languages?”**
LING 114: How to Learn a Foreign Language, University of Arizona, March 14
- 2025 **“brms tutorial.”**
LING 217: Experimental Phonology, UCLA, Nov. 18
- 2025 **“Origins and nature of food vocabulary in English.”**
Classics 185: Origins and Nature of English Vocabulary, UCLA, Jul. 8.
- 2025 **“Historical linguistics.”**
LING-20: Introduction to the Study of Language, UCLA (with Aidan Holmgren and Nick Guymon).
- 2024 **“Karma is the breeze in my hair on the weekend: Sanskrit loanwords into English.”**
Classics 185: Origins and Nature of English Vocabulary, UCLA, Jul. 11.

Teaching

Spring 2026	Teaching Assistant LING-110: Introduction to Historical Linguistics, UCLA Instructor of Record: Dr. Thomas Motter
2026	Teaching Assistant CLS 52: Greco-Roman Technologies: Materials, Monuments, Machines Faculty Supervisor: Prof Sarah Beckmann
2025	Teaching Assistant CLS 51: Art and Archaeology of Ancient Greece Instructor of Record: Prof. John Papadopoulos
2025	Instructor of Record LING-110: Introduction to Historical Linguistics, UCLA Faculty Supervisor: Prof. Stefan Keine
2025	Undergraduate Supervisor LING-197: Infants' Discrimination of Phonetic Contrasts, UCLA Faculty Supervisor: Prof. Megha Sundara
2025	Teaching Assistant LING-20: Introduction to Linguistic Analysis, UCLA Instructor of Record: Prof. Ben Eischens, Prof. Anoop Mahajan
Fall 2024	Teaching Assistant LING-110: Introduction to Historical Linguistics, UCLA Instructor of Record: Prof. David Goldstein
Sum. 2024	Teaching Assistant LING-1: Introduction to the Study of Language, UCLA Instructor of Record: Corrina Fuller
Fall 2021	Instructor ALS-102: Urdu, Grinnell College

Research Experience

2026–	Research Associate Center for Human-Environmental Research
2025	Research Assistant Supervisor: Prof. Brent Vine, UCLA
2023–	Research Assistant Supervisor: Prof. David Goldstein, UCLA
2020	Research Assistant Supervisor: Prof. Grant McCall, Tulane University

Scholarships & Awards

2026	Keck Summer Graduate Fellowship
2025	Graduate Summer Research Mentorship Award
2024 & 2025	Morvarid Guiv Graduate Fellowship in Zoroastrian Studies

2024	UCLA/Keck Humanistic Inquiry Research Award (shared with Tom Francis, Klaus Kim, and Laura McLean)
2024	Graduate Summer Research Mentorship Award
2023–24	Institute for the Study of Global Antiquity Fellowship
2023	Society for Classical Studies Lionel Pearson Fellowship
2022–23	Joan Palevsky Chair Post-Baccalaureate Scholarship in Classics
2022	CAMWS Award for Outstanding Accomplishment in Classical Studies
2022	Gus and Elfie Kalfass Prize for Excellence in German Studies
2021	Gerald W. LaLonde Fellowship in Classical Studies
2021	UC Berkeley Intensive Greek Workshop Fellowship
2019	Institute for Global Engagement Summer Language Study Scholarship

Languages

Ancient	Advanced: Ancient Greek, Hittite, Gothic, Latin, Vedic Intermediate: Middle High German, Old Persian, Avestan, Luwian
Modern	(Near-)Native: Hindi/Urdu, Panjabi, English; German (C1)

Technical Skills & Programming Languages

Microsoft Office Suite, NetLogo, RevBayes, Beast2.0, Python, L^AT_EX, Julia, Stan, R
R packages: `sp`, `sf`, `brms`, `lme4`, `mgcv`, `rstan`, `R-INLA`, `geostan`, `inlabru`, `cmdstanR` and many others

Service

2025–	Abstract Reviewer, Southern California Undergraduate Linguistics Conference
2024–	Graduate Seminar Organizer (with Aidan Holmgren), PIES, UCLA
2023–	Mentor, Asian and Asian American Classical Caucus
2023–	Organizer, UCLA Indo-European Conference
2023–24	Graduate Student Association Representative, UCLA Undergraduate Council
2021–22	Co-Chair, Grinnell College Classical Studies Student Education Policy Committee
2020–22	Co-Chair, Grinnell College German Studies Student Education Policy Committee

Professional Organizations

2025–	American Society for Premodern Asia
2025–	Linguistic Society of America
2025–	Acoustical Society of America
2023–	Indogermanische Gesellschaft
2022–	Society for Classical Studies
2022–	Classical Association of Midwest and South

References

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